

Program: Diploma in Civil and Environmental Engineering	
Course Code: 3358	Course Title: Civil Engineering Drafting Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Impart basic knowledge of the principles of building planning and drawing
- Enable the students to prepare submission drawings and service plans

Course Prerequisites:

Topic	Course code	Course name	Semester
Engineering Graphics		Engineering Graphics	1
Elementary Mathematics		Engineering mathematics	1&2
Building construction		Building materials and construction	3

Course Outcomes:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of basic principles of building drawing and to estimate the quantity of wood work and concrete (elements of buildings).	14	Applying
CO2	Apply principles of building planning and prepare line plans by following the building rules	14	Applying
CO3	Sketch service plans	8	Applying
CO4	Prepare submission drawings	5	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4			3				

3-Strongly mapped, 2-Moderately mapped,1-Weakly mapped

Course Outline

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of basic principles of building drawing and to estimate quantity of wood work and concrete.		
M1.01	Discuss the Conventions (as per IS 962) and symbols for different materials and Graphical symbols for Structural elements and sanitary and electrical installations	3	Understanding
M1.02	Sketch plan and section of Strip footing-Wall footing with step- RCC column footing and combined footing.	3	Applying
M1.03	Sketch plan and sectional elevation of Doglegged RCC stair-Quarter turn RCC stair	3	Applying
M1.04	Draw Plan of tile roof showing all members of gable and hipped end	3	Applying
M1.05	Draw Elevation of a Steel Truss	2	Understanding

CO2	Apply principles of building planning and prepare line plans following building rules		
M2.01	Discuss the Principles of planning for Residential and Public building- Aspect, Prospect, Orientation, Grouping, Privacy, Elegance, Flexibility, Circulation, Furniture requirements, Sanitation, Economy. Discuss the Introduction to Rules and bye-laws of sanctioning authorities for construction work (KMBR, KPBR, CRZ Act, NBC, KMMCR). Discuss Space requirement and norms for minimum dimension of different units in the residential and public buildings as per Kerala Building Rules. Define coverage, covered area, floor space index, Parking area, Ventilation, Headroom, Mezzanine floor, Setback, Corner Plot, Cul-de-sac, Small plots, Huts, High rise building. Discuss Provisions for Persons with Disability and Elderly Persons. Discuss Approvals from Town and Country Planning Department, Read and interpret readymade Architectural building drawing (To be procured from Architect, Planning Consultants, Planning Engineer)	2	Understanding
M2.02	Prepare single line sketch plans for single storey and double storey residential buildings, primary school building, public health centre building in a plot for specific plinth area as per NBC and KMBR	3	Applying
M2.03	Prepare plan, section and elevation for flat roof single storey and double storey residential building of minimum three rooms including water closet(WC), bath and staircase as per principles of planning from line sketch given.	3	Applying
M2.04	Sketch plan, section and elevation for public buildings such as School building/Restaurant/Library/Primary Health Centre as per principles of planning from line sketch given.	3	Applying
M2.05	Draw site plan showing proposed building-septic tank and sock pit -underground sump-compound wall-gate –boundary measurement-exterior open space-orientation-road etc. based on building rules	3	Applying

	Lab Test-I	2	
CO3	Sketch service plans		
M3.01	Sketch Plumbing layout showing water supply and sanitary fixtures of a residential building(one BHK Residential Building)	2	Applying
M3.02	Sketch Electrical layout showing electrical fixtures and control points with example (one BHK Residential Building)	2	Applying
M3.03	Sketch Fire safety plan and evacuation route (For one storey of a high rise building with typical floor plan)	2	Applying
M3.04	Sketch HVAC layout for one floor of a shopping mall	2	Understanding
CO4	Prepare submission drawings of buildings		
M4.01	Open ended experiment	5	Applying
	Lab Test II	2	

Text/Reference:

T/R	BookTitle/Author
T1	Shah.M.G.Kale,CM,Patki,S.Y.,BuildingDrawing,McgrawHillPublishing
T2	Dr.A Achuthan, Dr.Balagopal & TS Prabhu: Building Planning and Drawing; Spades publishers
R1	Malik and Mayo, Civil Engineering Drawing, Computech Publication Ltd
R2	Swamy, Kumara; Rao, N, Kameshwara, A ., Building Planning and Drawing, Charotar Publication, Anand.
R3	PunmiaB.C.,andJainA.K., Building Construction, Firewall Media.
R4	Mohan,G.MuthuSobha:PrinciplesofArchitecture,OxfordUniversityPress
R5	S P Deodar:Building Planning and Science; Khanna publisher
R6	IS962:Code of practice for Architectural and Building Drawings
R7	Kerala Municipal Building Rule
R8	Kerala Panchayath Building Rule
R9	National Building Code
R10	Kerala Minor Mineral Concession Rules
R11	Harmonized guidelines and space standards for barrier free built environment for persons with disability and elderly persons, 2016
R12	Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/ https://www.coursera.org/learn/construction-cost-estimating
3	https://swayam.gov.in/